Course Syllabus

Edit

Welcome to CSE 100: Advanced Data Structures!

This is a course in advanced data structure design, analysis, and implementation. You will use and implement abstract data types and data structures like (un)balanced trees, graphs, priority queues, and hash tables. You will also learn how to make sophisticated use of memory management, pointers, and recursion in programs, and analyze them with theoretical and practical performance analysis.

Instructor

- **Niema Moshiri** (Instructor)
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Class Sessions

- **Zoom Link (PSSs + Discussions; NOT Lab Hours)**
  - **https://ucsd.zoom.us/j/97657883638 (https://ucsd.zoom.us/j/97657883638)**
- **Problem Solving Sessions (MWF 2 PM)**
  - In-Person: **CENTR 115 (https://map.concept3d.com/?id=1005#!m/163044?share)**
- **Discussion Sections (Friday 3 PM)**
  - In-Person: **CENTR 115 (https://map.concept3d.com/?id=1005#!m/163044?share)**
- **Office Hours with Niema (Zoom)**
  - One-on-one, by appointment (email me: **niema@ucsd.edu (mailto:niema@ucsd.edu)**)

Grade Breakdown

Assessments are broken down into the following 3 categories:

1. **Preparation:** Reading Quizzes (RQs)
   - We will have a RQ due before every Lecture
   - Each RQ is **worth the same** (regardless of how many questions it has)
2. **Application:** Programming Assignments (PAs) + Projects
   - We will have **6 PAs**, each worth **10%** of the "Application" category
   - We will have **2 Projects**, each worth **20%** of the "Application" category
3. **Examination:** Midterm + Final
   - If you score **higher on the Midterm** than on the Final, **50% Midterm** and **50% Final**
   - If you score **higher on the Final** than on the Midterm, **25% Midterm** and **75% Final**

Your overall grade in the course will be the **minimum** of these 3 categories.

Example Grade Calculation

For example, if you score an average of 85% on your RQs, 100% on your PAs and Projects, 60% on the Midterm, and 80% on the Final:

- Your "Preparation" grade will be 85%
- Your "Application" grade will be 100%
- Your "Examination" grade will be (Midterm × 0.25) + (Final × 0.75) = (60% × 0.25) + (80% × 0.75) = 75%
- Your overall grade in the course will be min(Preparation, Application, Examination) = min(85%, 100%, 75%) = 75%

By default, we will use the standard scale of 90%—100% = A, 80%—89.9% = B, 70%—79.9% = C, 60%—69.9% = D, and <60% = F. These cutoffs may be lowered if need be, but they will never be raised. In other words, we may make it *easier* to get a certain letter grade, but never harder. Cutoffs for pluses and minuses will be determined by the overall grade distribution of the class.

Course Components

This course will be taught as a "flipped classroom," meaning you will learn the content *before* class via the assigned instructional materials, and during class, we will focus on addressing and correcting your misconceptions via problem solving and discussion. Generally, any given day of the course schedule (let's use Date X as an example), you will be expected to complete the following tasks:

- Complete the **Reading Assignments** and **Topic Videos** *before* Date X
- Complete the **Reading Quiz (RQ)** *before* Date X
- Attend the **Problem Solving Session (PSS)** *on* Date X
- Attend the **Discussion** *after* Date X
- Work on the current **Programming Assignment (PA)** or **Project** (and complete it before its deadline)

Before Class: Reading Assignments and Topic Videos

Content in the course will be delivered asynchronously in the form of **Reading Assignments** from this course's free online interactive textbook, ***Data Structures*** **(https://cogniterra.org/330)**, as well as short **Topic Videos** (~5–10 minutes each). Each day of the course schedule will have a few Reading Assignments and Topic Videos, and you are expected to go through them *before* the class on the corresponding day.

The interactive textbook has embedded multiple choice, free response, and even coding challenges, so you may wish to work through the challenges it contains for practice (though the challenges embedded throughout the textbook will not be factored into your course grade).

Before Class: Reading Quizzes (RQs)

Prior to a given day of the course schedule, after you have finished the Topic Videos (and potentially the optional readings) assigned for that day, you will complete a short **Reading Quiz (RQ)**, which will be due at **8:00 AM on that day**. All RQs are automatically graded instantly upon submission and will provide you feedback, and you have unlimited attempts until the deadline. **You are allowed to collaborate with your peers on the RQs.**

RQ points cannot be made up, no RQs will be dropped, and **late RQ submissions will not be accepted** in this course.

During Class: Problem Solving Sessions (PSSs)

Instead of holding traditional lecture-style sessions, we will be holding synchronous instructor-led **Problem Solving Sessions (PSSs)**. We expect that all students have at least been exposed to the topics for a given PSS *prior* to attending (via the corresponding topic videos, assigned readings, and RQ). Thus, rather than regurgitating the exact same information, these PSSs will focus on the following:

- **Asking Questions.** As you watch the topic videos or work on the RQs, you will likely be confused (which is a good thing!). During the PSSs, you will be able to ask any questions you may have about the topic, and we will work together to overcome the confusion.
- **Discussing Concepts.** The topic videos and readings are by no means comprehensive: they merely scratch the surface of a given concept. During the PSSs, we will explore the concepts more deeply as a group.
- **Solving Problems.** In this class, we will emphasize "learning-by-doing," and while the RQs will help you get a taste of the types of problems you may encounter, the PSSs will allow us to think about more challenging problems, with an emphasis on real-world scenarios.

PSS attendance will be **completely optional**, and recordings of the PSSs will be made available for you to follow along (or re-watch) asynchronously. However, we strongly encourage as much participation as possible, and all students will be expected to either attend or watch the recording of ***every*** PSS.

After Class: Discussion

Each week, we will be holding synchronous TA-led **Discussion** sections, which will focus on the following:

- **Starting the PAs/Projects.** We understand that one of the most daunting steps when coding is the very first step: How do I even start? During the Discussion sections, we will provide an overview of the current PA/Project and provide some tips regarding how to approach it.
- **Asking Questions.** You will likely encounter the majority of your "learning breakdowns" as you work on the PAs and Projects, which may occur after the corresponding PSS. During the Discussion sections, you are welcome (and encouraged!) to ask any questions you may have.

- **Solving Problems.** You will have already attempted some problems while attempting the RQs and working on the PSSs, but those problems may not have been closely tied to the PAs/Projects. During the Discussion sections, we will work on problems more strongly resembling what you may encounter on the PAs/Projects.

Discussion attendance will be **completely optional**, and recordings of the Discussion sections will be made available for you to follow along (or re-watch) asynchronously. However, we strongly encourage as much participation as possible, and all students will be expected to either attend or watch the recording of ***every*** Discussion.

Programming Assignments (PAs) and Projects

The **Programming Assignments (PAs)** will be the core form of assessment in this course. Typically, we will release a PA each **Wednesday**, which will be due at **10:00 PM** the following **Tuesday**.

Each PA will focus on the content covered in the 3 lectures during the week the PA was released. For example, if a PA is released on the Wednesday of Week 5, it will focus on content covered during the Monday, Wednesday, and Friday lectures of Week 5. All PAs are automatically graded instantly upon submission and will provide you feedback, and you have unlimited attempts until the deadline.

In addition to the PAs, which are generally split into smaller chunks, we will have two larger **Projects** towards the end of the course. These will combine data structures knowledge you acquire from class and homework with the pragmatics of writing a significant application that demands high performance.

For both, the grade you will receive is the grade of your highest-scoring on-time submission. PAs and Projects will only be graded based on correctness: we will not grade for programming style (e.g. comments, indentation, etc.). However, we strongly urge you to maintain good programming style as you work on the PAs and Projects, as clean code is easier to read, interpret, and ultimately debug. Also, **course staff will skip your ticket in the labs if they are unable to read your code easily**.

While we encourage discussing higher-level conceptual aspects of the PAs and Projects with your peers, **all work must be yours alone**. Sharing code (or anything easily translated into code) is a violation of UCSD's Academic Honesty policy and is not allowed.

Late PA and Project submissions will <u>not</u> be accepted in this course, and no PAs/Projects will be dropped. Start early, start often.

Exams

You will have two exams in this course: a Midterm and a Final. The exams will be administered asynchronously: you will have a larger window of time within which you can choose to start the exam at any time, and from the moment you start the exam, you will have a limited amount of time to complete the exam. Exams will be self-scheduled and administered by the Triton Testing Center (TTC) in the

Computer-Based Testing Lab. The TTC's rules concerning testing are the rules for this course. Some specific important rules to be mindful of:

- A physical photo ID is required to take a test. While a UC San Diego student ID is preferred, the TTC will accept a government-issued ID such as a driver's license or passport.
- The only items allowed in the TTC are a photo ID and writing utensil. All other personal items must be stored in the lockers. This includes wallets, keys, lanyards, and water bottles.
- Check-in begins approximately 10 minutes before test time and tests start promptly at the scheduled time. No additional time can be given for late arrivals.

You must schedule your exams in advance, and it is recommended that you do so as soon as possible. Scheduling of the Midterm opens on the first day of instruction. To schedule, visit **PrairieTest (https://prairietest.com/)** and log in with your UCSD credentials. We are still determining the logistics of scheduling of the Final, and we will provide more details closer to Finals Week. More information about testing policies and procedures can be found on the **TTC's website (https://tritontesting.ucsd.edu/)**. You may also email **tritontesting@ucsd.edu (mailto:tritontesting@ucsd.edu)** for assistance.

Please note that, if you plan to use OSD-approved accommodations for your exam, there are special instructions for scheduling your accommodated exam. Please see the **TTC's accommodated exam instructions page (https://tritontesting.ucsd.edu/accommodated-and-makeup-testing/students/index.html)** for more information. As with general exam scheduling, please be sure to schedule your accommodated exam as soon as possible.

Academic Integrity

All students **must** submit the Academic Integrity Agreement at the start of the course. **You will not receive <u>any</u> course credit until you submit the form.** By submitting the form, you are agreeing to its terms, so be sure to read it carefully. If you are found to have violated the course's Academic Integrity Policy, you will receive an **automatic F in the course**, and you may face even stricter sanctions from the University. In short, **do not cheat!**

Getting Help

We expect that *all* students will need help at some point in this course. If you find yourself needing help, this is not cause for embarrassment: it is completely expected, and our goal is to ensure that you are able to receive the help you need. Please be sure to seek help early and often through any (or all!) of the following resources:

Lab Hours

There are many, many virtual Lab Hours in which course staff are willing and available to help you with any questions you might have. Lab Hours are the only place where you should seek help with PAs specifically (though you are also welcome to come to Lab Hours for non-PA-related questions).

When you submit a ticket to the Lab Queue, you will include a Zoom link, and a member of the course staff will then accept your ticket and join your Zoom call. They will help you with your issue, and then they will leave the meeting and resolve the ticket.

- You can, for example, start a screen share with your code to help them immediately see your issue and present what you're seeing.
- Please refer to the **UCSD Zoom: User Guide (https://blink.ucsd.edu/technology/file-sharing/zoom/guide/index.html)** for detailed functionality instructions, including how to **Share Screen (https://support.ucsd.edu/its?id=kb_article_view&sysparm_article=KB0020142)** and **Annotate Shared Content (https://support.ucsd.edu/its?id=kb_article_view&sysparm_article=KB0020142#annotate)** .

Discussion Board

Please use this Discussion Board for any questions related to the RQs, PAs, course logistics, etc.

- When you create a new post, you will have the ability to specify the specific category that best fits your post.
- The Discussion Board is **not** for thorough/lengthy code-specific help: if you need help debugging your code, please seek help in Lab Hours!

Thank you for reading the course syllabus! Niema's favorite video game is Final Fantasy VII.